COPYRIGHT RESERVED **BCA (Sem-I) – Mathematics**

(103) GE-I

2025

Time : $1\frac{1}{2}$ hours

Full Marks : 40

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section–A

1. Answer any **four** questions of the following : 5×4=20

(a) State and prove Euler's theorem on homogeneous function of two variables.

(b) State and prove Taylor's theorem with Lagrange's form of remainder.

(c) Find the maximum and minimum values of $9x^2+12x+2, x\neq 0, y\neq 0$.

(d) Establish a formula for the area bounded by the curve $y=f(x)$, the x-axis and the two ordinates $x=a$ and $y=b$.

(e) Prove that $\int_0^a f(x)dx=\int_0^a f(x-a)dx$.

(f) Explain the method of finding the solution of homogeneous differential equation.

(g) Explain the method of finding the complementary function (C.F.) of a differential equation with constant co-efficients.

Section–B

2. Answer any **four** questions of the following : 5×4=20

(a) If $y=e^{a\sin^{-1}x}$, prove that

$$(1-x^2)y_2-xy_1-a^2y=0$$

(b) If $u=\sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, prove that

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\tan u$$

(c) Apply Maclaurin's theorem prove that

$$\sin x=x-\frac{x^3}{\lfloor 3}+\frac{x^5}{\lfloor 5}+\ldots.+\frac{x^n}{\lfloor n}\sin\frac{n\pi}{2}+\ldots.$$

(d) Integrate $\int_0^{\pi/2}\frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx$

(e) Find the area of the curve $x^{\frac{2}{3}}+y^{\frac{2}{3}}=a^{\frac{2}{3}}$

(f) Solve $\frac{dy}{dx}+\frac{1}{x}y=x^3$

(g) Solve $\frac{d^2y}{dx^2}+3\frac{dy}{dx}+2y=e^{2x}$

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